

Elearning Purchase Database Analysis

Tasks

1. Database Setup & Data Entry

- Create a new database.
- Create the three tables with proper:
 - Primary keys, Foreign keys,
 - Appropriate data types
- Insert sample data:
 - 4–5 learners
 - 4–5 courses (different categories)
 - 6–8 purchase records

Learners Table:

learner_id	full_name	country
1	Anitha Raj	India
2	Priya Charan	India
3	Divya Priya	India
4	Vijay Anand	India
5	Sanjay Kumar	India
6	Swetha Rani	USA
7	Manoj Kumar	USA
8	Deepa Lakshmi	USA
9	Michael Brown	UK
10	Emma Johnson	UK
11	Oliver Jones	Canada
12	Sophia Wilson	Canada
NULL	NULL	NULL

Courses Table:

course_id	course_name	category	unit_price
101	Excel for Beginners	Beginner	2500.00
102	Python Basics	Beginner	3000.00
103	SQL for Beginners	Beginner	2800.00
104	HTML and CSS	Web Development	6000.00
105	JavaScript Fundamentals	Web Development	7000.00
106	React JS Development	Web Development	8000.00
107	Java Programming	Programming	6500.00
108	C/C++ Programming	Programming	7500.00
109	AWS Cloud Fundamentals	Cloud Computing	8000.00
110	Microsoft Azure Basics	Cloud Computing	8000.00
111	Network Security	Cybersecurity	7500.00
112	Cybersecurity Fundamen...	Cybersecurity	8000.00
113	Digital Marketing Basics	Digital Marketing	4000.00
114	Social Media Marketing	Digital Marketing	6000.00
115	UI/UX Design Fundament...	UI/UX Design	3500.00
116	Graphic Design Basics	UI/UX Design	4000.00
NULL	NULL	NULL	NULL

Purchases Table:

purchase_id	learner_id	course_id	quantity	purchase_date
1001	1	101	1	2026-01-05
1002	1	105	2	2026-01-12
1003	1	107	1	2026-02-10
1004	2	102	1	2026-01-08
1005	2	106	1	2026-02-15
1006	2	112	1	2026-03-05
1007	3	103	2	2026-01-15
1008	3	104	1	2026-02-20
1009	4	105	1	2026-01-20
1010	4	110	1	2026-03-01
1011	4	113	1	2026-03-15
1012	5	101	2	2026-01-25
1013	5	108	1	2026-02-25
1014	6	106	1	2026-01-10
1015	6	111	1	2026-02-10
1016	7	105	1	2026-01-18
1017	7	110	2	2026-02-18
1018	8	102	1	2026-01-22
1019	8	109	1	2026-03-10
1020	9	107	1	2026-01-12
1021	9	113	1	2026-02-12
1022	10	103	1	2026-01-28

2. Data Exploration Using Joins Use INNER, LEFT, and RIGHT JOIN to:

- Combine learner, course, and purchase data
- Display: Learner name, Course name, Category, Quantity, Total amount, Purchase date
- Presentation Requirements:
 - Format currency to 2 decimal places
 - Use column aliases
 - Sort by the highest total amount

A) INNER JOIN--Displays only purchases where the learners and courses exist

full_name	course_name	category	Total_Amount	purchase_date
Manoj Kumar	Microsoft Azure Basics	Cloud Computing	16000.00	2026-02-18
Anitha Raj	JavaScript Fundamentals	Web Development	14000.00	2026-01-12
Anitha Raj	React JS Development	Web Development	8000.00	2026-04-01
Priya Charan	React JS Development	Web Development	8000.00	2026-02-15
Priya Charan	Cybersecurity Fundamentals	Cybersecurity	8000.00	2026-03-05
Vijay Anand	Microsoft Azure Basics	Cloud Computing	8000.00	2026-03-01
Swetha Rani	React JS Development	Web Development	8000.00	2026-01-10
Manoj Kumar	React JS Development	Web Development	8000.00	2026-04-10
Deepa Lakshmi	AWS Cloud Fundamentals	Cloud Computing	8000.00	2026-03-10
Emma Johnson	Cybersecurity Fundamentals	Cybersecurity	8000.00	2026-02-28
Oliver Jones	Microsoft Azure Basics	Cloud Computing	8000.00	2026-03-20
Vijay Anand	Network Security	Cybersecurity	7500.00	2026-04-05
Sanjay Kumar	C, C++ Programming	Programming	7500.00	2026-02-25
Swetha Rani	Network Security	Cybersecurity	7500.00	2026-02-10
Oliver Jones	C, C++ Programming	Programming	7500.00	2026-01-30
Vijay Anand	JavaScript Fundamentals	Web Development	7000.00	2026-01-20
Manoj Kumar	JavaScript Fundamentals	Web Development	7000.00	2026-01-18
Anitha Raj	Java Programming	Programming	6500.00	2026-02-10
Michael Brown	Java Programming	Programming	6500.00	2026-01-12
Divya Priya	HTML and CSS	Web Development	6000.00	2026-02-20
Sophia Wilson	HTML and CSS	Web Development	6000.00	2026-03-05
Divya Priya	SQL for Beginners	Beginner	5600.00	2026-01-15

B) LEFT JOIN-- Displays all learners, even if a learner has no purchase

	full_name	course_name	category	Total_Amount	purchase_date
▶	Manoj Kumar	Microsoft Azure Basics	Cloud Computing	16000.00	2026-02-18
	Anitha Raj	JavaScript Fundamentals	Web Development	14000.00	2026-01-12
	Anitha Raj	React JS Development	Web Development	8000.00	2026-04-01
	Priya Charan	React JS Development	Web Development	8000.00	2026-02-15
	Priya Charan	Cybersecurity Fundamentals	Cybersecurity	8000.00	2026-03-05
	Vijay Anand	Microsoft Azure Basics	Cloud Computing	8000.00	2026-03-01
	Swetha Rani	React JS Development	Web Development	8000.00	2026-01-10
	Manoj Kumar	React JS Development	Web Development	8000.00	2026-04-10
	Deepa Lakshmi	AWS Cloud Fundamentals	Cloud Computing	8000.00	2026-03-10
	Emma Johnson	Cybersecurity Fundamentals	Cybersecurity	8000.00	2026-02-28
	Oliver Jones	Microsoft Azure Basics	Cloud Computing	8000.00	2026-03-20
	Vijay Anand	Network Security	Cybersecurity	7500.00	2026-04-05
	Sanjay Kumar	C,C++ Programming	Programming	7500.00	2026-02-25
	Swetha Rani	Network Security	Cybersecurity	7500.00	2026-02-10
	Oliver Jones	C,C++ Programming	Programming	7500.00	2026-01-30
	Vijay Anand	JavaScript Fundamentals	Web Development	7000.00	2026-01-20
	Manoj Kumar	JavaScript Fundamentals	Web Development	7000.00	2026-01-18
	Anitha Raj	Java Programming	Programming	6500.00	2026-02-10
	Michael Brown	Java Programming	Programming	6500.00	2026-01-12
	Divya Priya	HTML and CSS	Web Development	6000.00	2026-02-20
	Sophia Wilson	HTML and CSS	Web Development	6000.00	2026-03-05
	Divya Priya	SQL for Beginners	Beginner	5600.00	2026-01-15

C) RIGHT JOIN-- Display all courses, even if a course has never been purchased

	full_name	course_name	category	Total_Amount	purchase_date
	Sanjay Kumar	C,C++ Programming	Programming	7500.00	2026-02-25
	Oliver Jones	C,C++ Programming	Programming	7500.00	2026-01-30
	Swetha Rani	Network Security	Cybersecurity	7500.00	2026-02-10
	Vijay Anand	Network Security	Cybersecurity	7500.00	2026-04-05
	Vijay Anand	JavaScript Fundamentals	Web Development	7000.00	2026-01-20
	Manoj Kumar	JavaScript Fundamentals	Web Development	7000.00	2026-01-18
	Anitha Raj	Java Programming	Programming	6500.00	2026-02-10
	Michael Brown	Java Programming	Programming	6500.00	2026-01-12
	Divya Priya	HTML and CSS	Web Development	6000.00	2026-02-20
	Sophia Wilson	HTML and CSS	Web Development	6000.00	2026-03-05
	Divya Priya	SQL for Beginners	Beginner	5600.00	2026-01-15
	Sanjay Kumar	Excel for Beginners	Beginner	5000.00	2026-01-25
	Vijay Anand	Digital Marketing Basics	Digital Marketing	4000.00	2026-03-15
	Michael Brown	Digital Marketing Basics	Digital Marketing	4000.00	2026-02-12
	Priya Charan	Python Basics	Beginner	3000.00	2026-01-08
	Deepa Lakshmi	Python Basics	Beginner	3000.00	2026-01-22
	Emma Johnson	SQL for Beginners	Beginner	2800.00	2026-01-28
	Anitha Raj	Excel for Beginners	Beginner	2500.00	2026-01-05
	Sophia Wilson	Excel for Beginners	Beginner	2500.00	2026-02-05
	NULL	Social Media Marketing	Digital Marketing	NULL	NULL
	NULL	UI/UX Design Fundamentals	UI/UX Design	NULL	NULL
	NULL	Graphic Design Basics	UI/UX Design	NULL	NULL

3. Core Analytical Queries (Q1–Q5)

Q1. Display each learner's total spending with their country

learner_id	full_name	country	Total_Spending
▶ 1	Anitha Raj	India	31000.00
2	Priya Charan	India	19000.00
3	Divya Priya	India	11600.00
4	Vijay Anand	India	26500.00
5	Sanjay Kumar	India	12500.00
6	Swetha Rani	USA	15500.00
7	Manoj Kumar	USA	31000.00
8	Deepa Lakshmi	USA	11000.00
9	Michael Brown	UK	10500.00
10	Emma Johnson	UK	10800.00
11	Oliver Jones	Canada	15500.00
12	Sophia Wilson	Canada	8500.00

Q2. Find the top 3 most purchased courses by quantity

course_id	course_name	Total_Quantity
101	Excel for Beginners	4
105	JavaScript Fundamentals	4
106	React JS Development	4

Q3. Show each category's: • Total revenue • Number of unique learners

category	Total_Revenue	Unique_Learners
Beginner	24400.00	7
Cloud Computing	40000.00	4
Cybersecurity	31000.00	4
Digital Marketing	8000.00	2
Programming	28000.00	4
Web Development	72000.00	7

Q4. List learners who purchased from more than one category.

learner_id	full_name	category_count
1	Anitha Raj	4
2	Priya Charan	3
3	Divya Priya	2
4	Vijay Anand	4
5	Sanjay Kumar	2
6	Swetha Rani	2
7	Manoj Kumar	3
8	Deepa Lakshmi	2
9	Michael Brown	2
10	Emma Johnson	2
11	Oliver Jones	2
12	Sophia Wilson	2

Q5. Identify courses never purchased.

learner_id	full_name	course_name	category
NULL	NULL	Social Media Marketing	Digital Marketing
NULL	NULL	UI/UX Design Fundamentals	UI/UX Design
NULL	NULL	Graphic Design Basics	UI/UX Design

4. Subqueries & Correlated Subqueries

Q6. Find learners whose total spending is above the average learner spending.

learner_id	full_name	Total_Spending2
1	Anitha Raj	31000.00
2	Priya Charan	19000.00
4	Vijay Anand	26500.00
7	Manoj Kumar	31000.00

Q7. Display courses whose price is higher than any course in the 'Beginner' category.

course_id	course_name	category	unit_price
104	HTML and CSS	Web Development	6000.00
105	JavaScript Fundamentals	Web Development	7000.00
106	React JS Development	Web Development	8000.00
107	Java Programming	Programming	6500.00
108	C,C++ Programming	Programming	7500.00
109	AWS Cloud Fundamentals	Cloud Computing	8000.00
110	Microsoft Azure Basics	Cloud Computing	8000.00
111	Network Security	Cybersecurity	7500.00
112	Cybersecurity Fundamentals	Cybersecurity	8000.00
113	Digital Marketing Basics	Digital Marketing	4000.00
114	Social Media Marketing	Digital Marketing	6000.00
115	UI/UX Design Fundamentals	UI/UX Design	3500.00
116	Graphic Design Basics	UI/UX Design	4000.00
NULL	NULL	NULL	NULL

Q8 . Find learners who spent more than the average spending in their country.

learner_id	full_name	country	Total_Spending
1	Anitha Raj	India	31000.00
4	Vijay Anand	India	26500.00
7	Manoj Kumar	USA	31000.00
10	Emma Johnson	UK	10800.00
11	Oliver Jones	Canada	15500.00

5. CTE, CASE, View, and NULL Handling

Q9. Use a CTE to calculate total spending per learner, then: Display learners with spending above 10,000.

full_name	Total_spending
Anitha Raj	31000.00
Priya Charan	19000.00
Divya Priya	11600.00
Vijay Anand	26500.00
Sanjay Kumar	12500.00
Swetha Rani	15500.00
Manoj Kumar	31000.00
Deepa Lakshmi	11000.00
Michael Brown	10500.00
Emma Johnson	10800.00
Oliver Jones	15500.00

Q10. CASE Expression Classify learners based on spending: • Above 15,000 → “High Value”, • 8,000–15,000 → “Medium Value”, • Below 8,000 → “Low Value”

learner_id	full_name	Total_Spending	Totalspending_category
1	Anitha Raj	31000.00	High Value
2	Priya Charan	19000.00	High Value
3	Divya Priya	11600.00	Medium Value
4	Vijay Anand	26500.00	High Value
5	Sanjay Kumar	12500.00	Medium Value
6	Swetha Rani	15500.00	High Value
7	Manoj Kumar	31000.00	High Value
8	Deepa Lakshmi	11000.00	Medium Value
9	Michael Brown	10500.00	Medium Value
10	Emma Johnson	10800.00	Medium Value
11	Oliver Jones	15500.00	High Value
12	Sophia Wilson	8500.00	Medium Value

Q11 . NULL Handling • Display all courses and replace NULL purchase counts with 0 using: IFNULL() or COALESCE()

course_id	course_name	category	Purchase_count
101	Excel for Beginners	Beginner	3
102	Python Basics	Beginner	2
103	SQL for Beginners	Beginner	2
104	HTML and CSS	Web Development	2
105	JavaScript Fundamentals	Web Development	3
106	React JS Development	Web Development	4
107	Java Programming	Programming	2
108	C,C++ Programming	Programming	2
109	AWS Cloud Fundamentals	Cloud Computing	1
110	Microsoft Azure Basics	Cloud Computing	3
111	Network Security	Cybersecurity	2
112	Cybersecurity Fundamen...	Cybersecurity	2
113	Digital Marketing Basics	Digital Marketing	2
114	Social Media Marketing	Digital Marketing	0
115	UI/UX Design Fundament...	UI/UX Design	0
116	Graphic Design Basics	UI/UX Design	0

Q12 . View • Create a view: category_performance_view • Showing: • Category • Total revenue • Number of purchases • Average revenue per purchase

category	Total_revenue	Number_of_purchases	Average_revenue_per_purchase
Beginner	24400.00	7	3485.71
Web Development	72000.00	9	8000.00
Programming	28000.00	4	7000.00
Cloud Computing	40000.00	4	10000.00
Cybersecurity	31000.00	4	7750.00
Digital Marketing	8000.00	2	4000.00
UI/UX Design	0.00	0	0.00